

Jasroop Singh Chawla

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EDUCATION

Thapar Institute of Engineering and Technology, Patiala

2022 – 2026

Bachelor of Engineering in Computer Science and Engineering

CGPA: 8.39/10

The Senior Study School II, Amritsar

Class XII – 97.4%

2020 – 2022

Class X – 89.2%

2019 – 2020

SKILLS

- **Programming:** Python, SQL, C++, Shell
- **ML/DS:** Scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Optuna
- **Deep Learning:** PyTorch, TensorFlow, Keras, Hugging Face Transformers
- **NLP/LLMs:** BERT, GPT, LangChain, RAG, Prompt Engineering
- **Computer Vision:** OpenCV, CNNs, Transfer Learning, ViT, YOLO
- **MLOps/Cloud:** MLflow, DVC, Docker, AWS S3, Azure, GCP
- **Deployment:** FastAPI, Flask, Streamlit, Gradio, CI/CD
- **Databases:** PostgreSQL, MongoDB, BigQuery
- **Visualization:** Seaborn, Plotly, Power BI, Tableau
- **Tools:** Git, Jupyter, VS Code, Google Colab

PROJECTS

1. Autonomous Research Agent using LangChain + OpenAI

LLMs, LangChain, GPT-4

Developed an autonomous multi-step agent capable of answering complex user queries by planning sub-tasks, retrieving data from the web, summarizing PDFs, and chaining LLM calls. Integrated memory, tools, and search APIs to enable reasoning and execution.

2. Multi-Modal Emotion Classification

PyTorch, ViT, BERT, Fusion Models

Built a multi-modal model that classifies human emotion by combining facial expressions (images) and spoken text. Used pretrained Vision Transformer and BERT, and implemented early and late fusion strategies to enhance accuracy across real-world datasets.

3. End-to-End ML Pipeline with Feature Store and MLflow

Scikit-learn, MLflow, FastAPI

Implemented a complete ML lifecycle: data ingestion, feature engineering, model training, tracking, and API deployment. Integrated a feature store with automated logging (MLflow), versioned models, and deployed a FastAPI endpoint for real-time predictions.

PUBLICATIONS & CERTIFICATIONS

Conference Paper: *Tomato Classification Using Deep Learning Techniques*

Presented at EAIC 2025, NIT Jalandhar – Custom CNN model for tomato disease detection, compared with transfer learning models.

Certifications (Coursera):

- Deep Learning Specialization (Andrew Ng)
- Natural Language Processing with Classification and Vector Spaces
- Machine Learning with Python
- Generative AI with LLMs and Prompt Engineering
- [Full list of certifications and projects](#)

LEADERSHIP & ROLES

Documentation Head – Youth United, Thapar Chapter

2021 – Present

Led documentation and reporting for events promoting youth leadership and community development.

Discipline Roles – Thapar Institute & Senior Study School

Held leadership roles in hostel discipline committees and student councils; supported institutional governance and event execution.